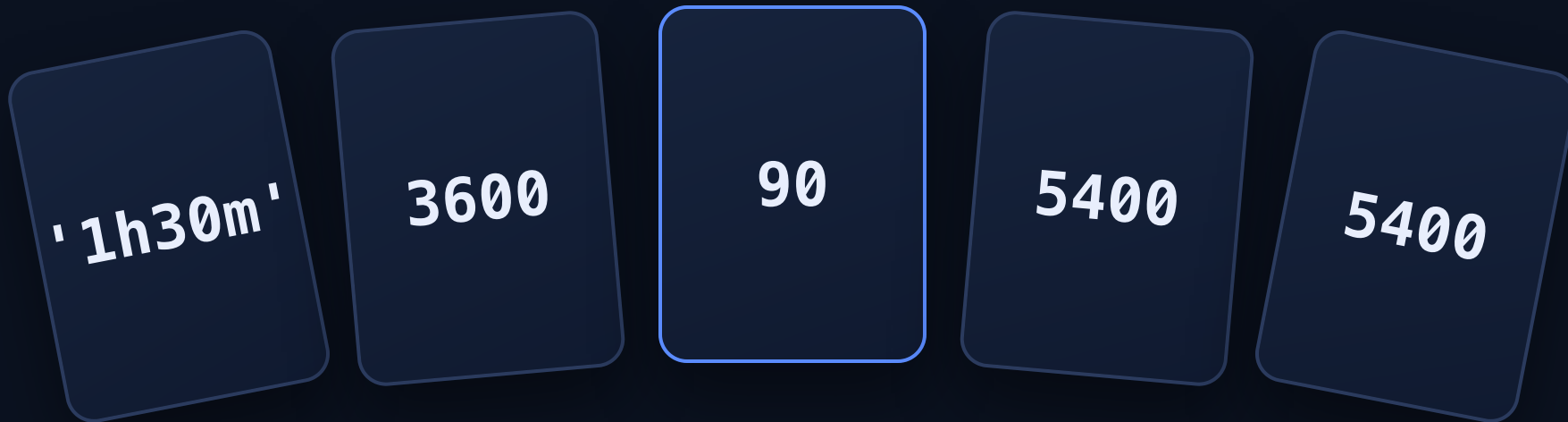


IBM HACKATHON · BOB IDE

RASHOMON

unit tests for readability



five readers · one answer · only the code decides

PROBLEM

We measure the file, not the reader

58% of development time is spent reading code — yet 121 readability metrics count tokens, lines and complexity, and were shown **not to track human judgement**.

Asking an LLM “is this readable?” correlates with humans at roughly $\rho \approx 0$.

INSIGHT

Don't ask. Check.

Ask what the function **returns** — then **run it**. A wrong prediction is ground truth that the source misled a reader.

Five readers who disagree give you a **distribution**, not an opinion.

HOW IT WORKS

Five isolated readers, one packet

- Bob fans out five **explore** subagents — **fork_context: false**, one tool call each
- **.bobignore** hides answers and the target repo; a sealed canary proves it
- The harness executes probes, grades predictions, and **git log** supplies the fix history

RESULTS — LIVE ON 30 BOLTONS FUNCTIONS

410 predictions, graded by execution

9

Consensus misread
(every reader wrong)

4

Scattered
(readers disagree)

17

Clear
(all readers clean)

0.394

ρ vs bug-fix history
 $p = 0.031$, CI [0.03, 0.71]

On the demo: **parse_duration** misleads 7 of 15 readings while **mean** is read by all five — with no human label anywhere.

PR GATE

Readability you can regress on

make gate writes a pull-request comment with a misread rate per function — and fails **only** when a change makes code more confusing.

A check a team can run on every PR, not a review argument. The Clarifier rewrites the confusing code and **equiv** proves behaviour didn't move.

MARKET & COMPETITION

The only one that runs the answer

- **CoReEval** asks an LLM if code is readable — $p \approx 0$ against humans
- **CRUXEval** uses code to grade *models* — Rashomon uses models to grade *code*
- Linters count tokens; nobody counts **readings**

Independent readers + execution + git history = a **test**.

FUTURE

Where it goes next

- JS / TS targets and editor squiggles on confusing code
- Cross-model-family witness panel (Gemini already wired; watsonx pending)
- A sixth independent reader on watsonx.ai

